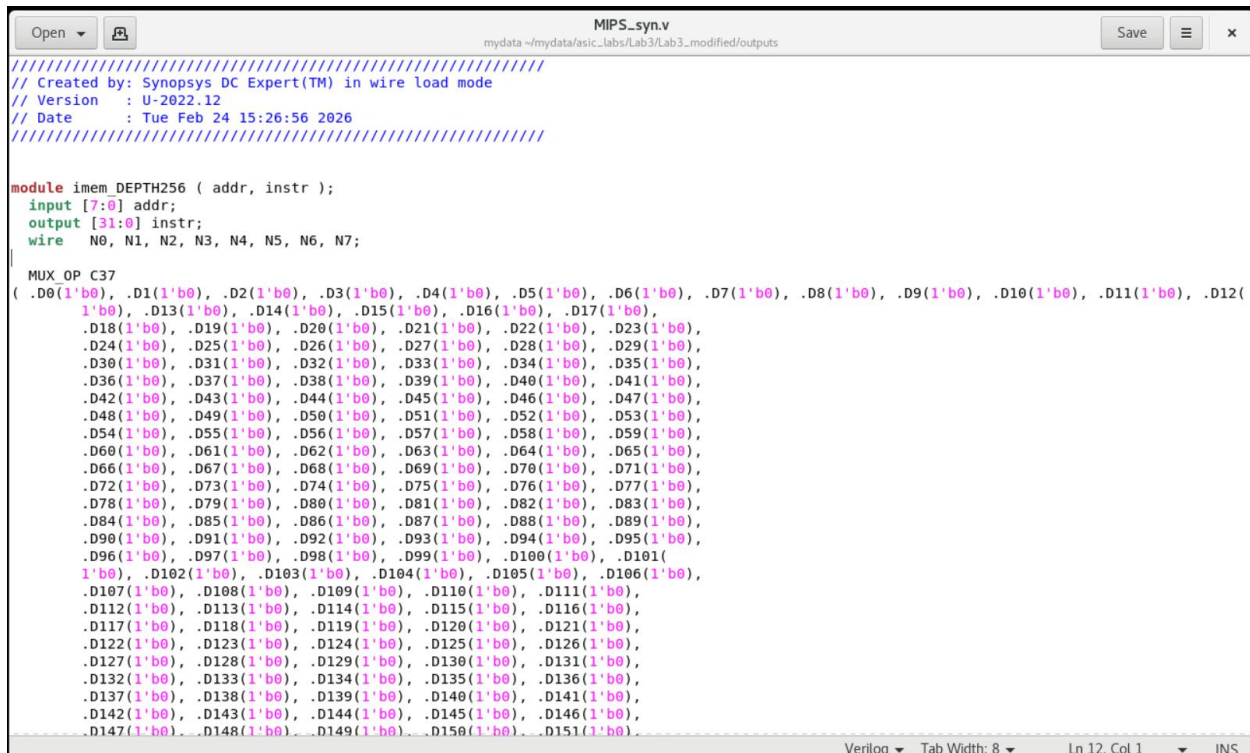

Lab 3 – Logical Synthesis

Lab 3 – COMPENG 4DV4

Chris Hoftyzer 400410904 - Group
12

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Checkpoint 1:



```
Open  MIPS_syn.v
mydata ~/mydata/asic_labs/Lab3/Lab3_modified/outputs Save

/////////////////////////////////////////////////////////////////
// Created by: Synopsys DC Expert(TM) in wire load mode
// Version   : U-2022.12
// Date      : Tue Feb 24 15:26:56 2026
/////////////////////////////////////////////////////////////////

module imem_DEPTH256 ( addr, instr );
  input [7:0] addr;
  output [31:0] instr;
  wire  N0, N1, N2, N3, N4, N5, N6, N7;

  MUX_OP C37
  ( .D0(1'b0), .D1(1'b0), .D2(1'b0), .D3(1'b0), .D4(1'b0), .D5(1'b0), .D6(1'b0), .D7(1'b0), .D8(1'b0), .D9(1'b0), .D10(1'b0), .D11(1'b0), .D12(
    1'b0), .D13(1'b0), .D14(1'b0), .D15(1'b0), .D16(1'b0), .D17(1'b0),
    .D18(1'b0), .D19(1'b0), .D20(1'b0), .D21(1'b0), .D22(1'b0), .D23(1'b0),
    .D24(1'b0), .D25(1'b0), .D26(1'b0), .D27(1'b0), .D28(1'b0), .D29(1'b0),
    .D30(1'b0), .D31(1'b0), .D32(1'b0), .D33(1'b0), .D34(1'b0), .D35(1'b0),
    .D36(1'b0), .D37(1'b0), .D38(1'b0), .D39(1'b0), .D40(1'b0), .D41(1'b0),
    .D42(1'b0), .D43(1'b0), .D44(1'b0), .D45(1'b0), .D46(1'b0), .D47(1'b0),
    .D48(1'b0), .D49(1'b0), .D50(1'b0), .D51(1'b0), .D52(1'b0), .D53(1'b0),
    .D54(1'b0), .D55(1'b0), .D56(1'b0), .D57(1'b0), .D58(1'b0), .D59(1'b0),
    .D60(1'b0), .D61(1'b0), .D62(1'b0), .D63(1'b0), .D64(1'b0), .D65(1'b0),
    .D66(1'b0), .D67(1'b0), .D68(1'b0), .D69(1'b0), .D70(1'b0), .D71(1'b0),
    .D72(1'b0), .D73(1'b0), .D74(1'b0), .D75(1'b0), .D76(1'b0), .D77(1'b0),
    .D78(1'b0), .D79(1'b0), .D80(1'b0), .D81(1'b0), .D82(1'b0), .D83(1'b0),
    .D84(1'b0), .D85(1'b0), .D86(1'b0), .D87(1'b0), .D88(1'b0), .D89(1'b0),
    .D90(1'b0), .D91(1'b0), .D92(1'b0), .D93(1'b0), .D94(1'b0), .D95(1'b0),
    .D96(1'b0), .D97(1'b0), .D98(1'b0), .D99(1'b0), .D100(1'b0), .D101(
    1'b0), .D102(1'b0), .D103(1'b0), .D104(1'b0), .D105(1'b0), .D106(1'b0),
    .D107(1'b0), .D108(1'b0), .D109(1'b0), .D110(1'b0), .D111(1'b0),
    .D112(1'b0), .D113(1'b0), .D114(1'b0), .D115(1'b0), .D116(1'b0),
    .D117(1'b0), .D118(1'b0), .D119(1'b0), .D120(1'b0), .D121(1'b0),
    .D122(1'b0), .D123(1'b0), .D124(1'b0), .D125(1'b0), .D126(1'b0),
    .D127(1'b0), .D128(1'b0), .D129(1'b0), .D130(1'b0), .D131(1'b0),
    .D132(1'b0), .D133(1'b0), .D134(1'b0), .D135(1'b0), .D136(1'b0),
    .D137(1'b0), .D138(1'b0), .D139(1'b0), .D140(1'b0), .D141(1'b0),
    .D142(1'b0), .D143(1'b0), .D144(1'b0), .D145(1'b0), .D146(1'b0),
    .D147(1'b0), .D148(1'b0), .D149(1'b0), .D150(1'b0), .D151(1'b0)
  );
endmodule
```

Figure 1 – MIPS_syn.v contents

Checkpoint 2:



The screenshot shows a text editor window titled "MIPS.sdf" with the file path "mydata ~/mydata/asic_labs/Lab3/Lab3_modified/outputs". The file contains a Verilog-like description of a MIPS processor. It starts with a DELAYFILE block containing metadata like SDFVERSION, DESIGN, DATE, and PROGRAM. This is followed by a CELL block with an INSTANCE of a DELAY block. The DELAY block contains an ABSOLUTE section with numerous INTERCONNECT statements defining the internal wiring of the processor, such as u_ctrl/C104/Z_0 to U373/A and u_dmem/C16717/Z_31 to U372/A1. The status bar at the bottom indicates "Plain Text", "Tab Width: 8", "Ln 1, Col 1", and "INS".

```
((DELAYFILE
(SDFVERSION "0VI 1.0")
(DESIGN "mips_cpu")
(DATE "Tue Feb 24 15:26:56 2026")
(VENDOR "asap7sc7p5t_A0_LVT_TT_nldm_211120")
(PROGRAM "Synopsys Design Compiler cmos")
(VERSION "U-2022.12")
(DIVIDER /)
(VOLTAGE 0.70:0.70:0.70)
(PROCESS)
(TEMPERATURE 25.00:25.00:25.00)
(TIMESCALE 1ps)
(CELL
  (CELLTYPE "mips_cpu")
  (INSTANCE)
  (DELAY
    (ABSOLUTE
      (INTERCONNECT u_ctrl/C104/Z_0 U373/A (0.000:0.000:0.000))
      (INTERCONNECT u_dmem/C16717/Z_31 U372/A1 (0.000:0.000:0.000))
      (INTERCONNECT u_ctrl/C107/Z_0 U372/A2 (0.000:0.000:0.000))
      (INTERCONNECT u_alu/U39/Y U372/B1 (0.000:0.000:0.000))
      (INTERCONNECT U343/Y U372/B2 (0.000:0.000:0.000))
      (INTERCONNECT u_dmem/C16717/Z_30 U371/A1 (0.000:0.000:0.000))
      (INTERCONNECT u_ctrl/C107/Z_0 U371/A2 (0.000:0.000:0.000))
      (INTERCONNECT u_alu/U31/Y U371/B1 (0.000:0.000:0.000))
      (INTERCONNECT U343/Y U371/B2 (0.000:0.000:0.000))
      (INTERCONNECT u_dmem/C16717/Z_29 U370/A1 (0.000:0.000:0.000))
      (INTERCONNECT u_ctrl/C107/Z_0 U370/A2 (0.000:0.000:0.000))
      (INTERCONNECT u_alu/U30/Y U370/B1 (0.000:0.000:0.000))
      (INTERCONNECT U343/Y U370/B2 (0.000:0.000:0.000))
      (INTERCONNECT u_dmem/C16717/Z_28 U369/A1 (0.000:0.000:0.000))
      (INTERCONNECT u_ctrl/C107/Z_0 U369/A2 (0.000:0.000:0.000))
      (INTERCONNECT u_alu/U53/Y U369/B1 (0.000:0.000:0.000))
      (INTERCONNECT U343/Y U369/B2 (0.000:0.000:0.000))
      (INTERCONNECT u_dmem/C16717/Z_27 U368/A1 (0.000:0.000:0.000))
      (INTERCONNECT u_ctrl/C107/Z_0 U368/A2 (0.000:0.000:0.000))
      (INTERCONNECT u_alu/U49/Y U368/B1 (0.000:0.000:0.000))
      (INTERCONNECT U343/Y U368/B2 (0.000:0.000:0.000))
      (INTERCONNECT u_dmem/C16717/Z_26 U367/A1 (0.000:0.000:0.000))
    )
  )
)
```

Figure 2 – MIPS.sdf contents

Checkpoint 3:



The screenshot shows a text editor window titled "MIPS.sdc" with the file path "mydata ~/mydata/asic_labs/Lab3/Lab3_modified/outputs". The file contains a Synopsys Design Compiler timing constraints file. It starts with a header line "# Created by write_sdc on Tue Feb 24 15:26:55 2026" and a separator line "#####". Below this, it sets the sdc_version to 2.1 and defines units for time (ps), resistance (kohm), capacitance (ff), voltage (V), and current (mA). It also sets the maximum area to 0 and creates a clock constraint for the 'clk' port with a period of 200 and a waveform of {0 100}. The status bar at the bottom indicates "Plain Text", "Tab Width: 8", "Ln 10, Col 42", and "INS".

```
#####
# Created by write_sdc on Tue Feb 24 15:26:55 2026
#####
set sdc_version 2.1

set_units -time ps -resistance kohm -capacitance ff -voltage V -current mA
set_max_area 0
create_clock [get_ports clk] -period 200 -waveform {0 100}
```

Figure 3 – MIPS.sdc contents

Checkpoint 4:

```

MIPS.area
mydata ~/mydata/asic_labs/Lab3/Lab3_modified/reports

Report : area
Design : mips_cpu
Version: U-2022.12
Date   : Tue Feb 24 15:26:55 2026

Library(s) Used:

  asap7sc7p5t_INVBUF_SLVT_TT_nldm_211120 (File: /home/v72180/mydata/asic_labs/Lab3/Lab3_modified/lib/db/
asap7sc7p5t_INVBUF_SLVT_TT_nldm_211120.db)
  asap7sc7p5t_A0_SLVT_TT_nldm_211120 (File: /home/v72180/mydata/asic_labs/Lab3/Lab3_modified/lib/db/asap7sc7p5t_A0_SLVT_TT_nldm_211120.db)
  asap7sc7p5t_OA_SLVT_TT_nldm_211120 (File: /home/v72180/mydata/asic_labs/Lab3/Lab3_modified/lib/db/asap7sc7p5t_OA_SLVT_TT_nldm_211120.db)
  asap7sc7p5t_SIMPLE_SLVT_TT_nldm_211120 (File: /home/v72180/mydata/asic_labs/Lab3/Lab3_modified/lib/db/
asap7sc7p5t_SIMPLE_SLVT_TT_nldm_211120.db)
  asap7sc7p5t_A0_LVT_TT_nldm_211120 (File: /home/v72180/mydata/asic_labs/Lab3/Lab3_modified/lib/db/asap7sc7p5t_A0_LVT_TT_nldm_211120.db)
  asap7sc7p5t_INVBUF_LVT_TT_nldm_211120 (File: /home/v72180/mydata/asic_labs/Lab3/Lab3_modified/lib/db/
asap7sc7p5t_INVBUF_LVT_TT_nldm_211120.db)
  asap7sc7p5t_SIMPLE_LVT_TT_nldm_211120 (File: /home/v72180/mydata/asic_labs/Lab3/Lab3_modified/lib/db/
asap7sc7p5t_SIMPLE_LVT_TT_nldm_211120.db)
  asap7sc7p5t_OA_LVT_TT_nldm_211120 (File: /home/v72180/mydata/asic_labs/Lab3/Lab3_modified/lib/db/asap7sc7p5t_OA_LVT_TT_nldm_211120.db)
  asap7sc7p5t_SEQ_SLVT_TT_nldm_220123 (File: /home/v72180/mydata/asic_labs/Lab3/Lab3_modified/lib/db/
asap7sc7p5t_SEQ_SLVT_TT_nldm_220123.db)
  asap7sc7p5t_SEQ_LVT_TT_nldm_220123 (File: /home/v72180/mydata/asic_labs/Lab3/Lab3_modified/lib/db/asap7sc7p5t_SEQ_LVT_TT_nldm_220123.db)
  gtech (File: /CMC/tools/synopsys/syn_vU-2022.12/syn/U-2022.12/libraries/syn/gtech.db)

Number of ports:          799
Number of nets:          12682
Number of cells:         11498
Number of combinational cells: 2213
Number of sequential cells:  9267
Number of macros/black boxes: 0
Number of buf/inv:        603
Number of references:      49

Combinational area:      136.468800
Buf/Inv area:            27.774900
Noncombinational area:   11.372400
  
```

Figure 4 – Mips.area contents

```

MIPS.timing
mydata ~/mydata/asic_labs/Lab3/Lab3_modified/reports

Information: Updating graph... (UID-83)
Warning: Design 'mips_cpu' contains 4 high-fanout nets. A fanout number of 1000 will be used for delay calculations involving these nets.
(TIM-134)
Information: Updating design information... (UID-85)
Warning: Design 'mips_cpu' contains 4 high-fanout nets. A fanout number of 1000 will be used for delay calculations involving these nets.
(TIM-134)

Report : timing
-path full
-delay max
-max_paths 1
Design : mips_cpu
Version: U-2022.12
Date   : Tue Feb 24 15:26:48 2026

# A fanout number of 1000 was used for high fanout net computations.

Operating Conditions: PVT_0P7V_25C Library: asap7sc7p5t_A0_LVT_TT_nldm_211120
Wire Load Model Mode: top

Startpoint: pc_q_reg[8]
(rising edge-triggered flip-flop clocked by clk)
Endpoint: pc_q_reg[2]
(rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

Point                                Incr      Path
-----
clock clk (rise edge)                 0.00      0.00
clock network delay (ideal)            0.00      0.00
pc_q_reg[8]/CLK (DFFASRHX1 ASAP7_75t_SL) 0.00 #    0.00 r
pc_q_reg[8]/QN (DFFASRHX1 ASAP7_75t_SL) 37.89     37.89 r
u_imem/addr[6] (imem_DEPTH256)         0.00      37.89 r
u_imem/B_6/Z (GTECH_BUF)               0.00      37.89 r
u_imem/C37/Z_0 (*MUX_OP_256_8_1)       0.00      37.89 r
u_imem/B_24/Z (GTECH_BUF)              0.00      37.89 r
  
```

Figure 5 – Mips.timing contents

```

MIPS.power
mydata ~/mydata/asic_labs/Lab3/Lab3_modified/reports
Save

Loading db file '/home/v72180/mydata/asic_labs/Lab3/Lab3_modified/lib/db/asap7sc7p5t_OA_SLVT_TT_nldm_211120.db'
Loading db file '/home/v72180/mydata/asic_labs/Lab3/Lab3_modified/lib/db/asap7sc7p5t_SEQ_LVT_TT_nldm_220123.db'
Loading db file '/home/v72180/mydata/asic_labs/Lab3/Lab3_modified/lib/db/asap7sc7p5t_SEQ_SLVT_TT_nldm_220123.db'
Loading db file '/home/v72180/mydata/asic_labs/Lab3/Lab3_modified/lib/db/asap7sc7p5t_SIMPLE_LVT_TT_nldm_211120.db'
Loading db file '/home/v72180/mydata/asic_labs/Lab3/Lab3_modified/lib/db/asap7sc7p5t_INVBUFF_LVT_TT_nldm_211120.db'
Loading db file '/home/v72180/mydata/asic_labs/Lab3/Lab3_modified/lib/db/asap7sc7p5t_INVBUFF_SLVT_TT_nldm_211120.db'
Loading db file '/home/v72180/mydata/asic_labs/Lab3/Lab3_modified/lib/db/asap7sc7p5t_A0_LVT_TT_nldm_211120.db'
Loading db file '/home/v72180/mydata/asic_labs/Lab3/Lab3_modified/lib/db/asap7sc7p5t_SIMPLE_SLVT_TT_nldm_211120.db'
Loading db file '/home/v72180/mydata/asic_labs/Lab3/Lab3_modified/lib/db/asap7sc7p5t_A0_SLVT_TT_nldm_211120.db'
Loading db file '/home/v72180/mydata/asic_labs/Lab3/Lab3_modified/lib/db/asap7sc7p5t_OA_LVT_TT_nldm_211120.db'
Information: Propagating switching activity (low effort zero delay simulation). (PWR-6)
Warning: Design has unannotated primary inputs. (PWR-414)
Warning: Design has unannotated sequential cell outputs. (PWR-415)
Warning: Design has unannotated black box outputs. (PWR-428)

*****
Report : power
        -analysis_effort low
Design : mips_cpu
Version: U-2022.12
Date   : Tue Feb 24 15:26:55 2026
*****

Library(s) Used:

    asap7sc7p5t_INVBUFF_SLVT_TT_nldm_211120 (File: /home/v72180/mydata/asic_labs/Lab3/Lab3_modified/lib/db/
asap7sc7p5t_INVBUFF_SLVT_TT_nldm_211120.db)
    asap7sc7p5t_A0_SLVT_TT_nldm_211120 (File: /home/v72180/mydata/asic_labs/Lab3/Lab3_modified/lib/db/asap7sc7p5t_A0_SLVT_TT_nldm_211120.db)
    asap7sc7p5t_OA_SLVT_TT_nldm_211120 (File: /home/v72180/mydata/asic_labs/Lab3/Lab3_modified/lib/db/asap7sc7p5t_OA_SLVT_TT_nldm_211120.db)
    asap7sc7p5t_SIMPLE_SLVT_TT_nldm_211120 (File: /home/v72180/mydata/asic_labs/Lab3/Lab3_modified/lib/db/
asap7sc7p5t_SIMPLE_SLVT_TT_nldm_211120.db)
    asap7sc7p5t_A0_LVT_TT_nldm_211120 (File: /home/v72180/mydata/asic_labs/Lab3/Lab3_modified/lib/db/asap7sc7p5t_A0_LVT_TT_nldm_211120.db)
    asap7sc7p5t_INVBUFF_LVT_TT_nldm_211120 (File: /home/v72180/mydata/asic_labs/Lab3/Lab3_modified/lib/db/
asap7sc7p5t_INVBUFF_LVT_TT_nldm_211120.db)
    asap7sc7p5t_SIMPLE_LVT_TT_nldm_211120 (File: /home/v72180/mydata/asic_labs/Lab3/Lab3_modified/lib/db/
asap7sc7p5t_SIMPLE_LVT_TT_nldm_211120.db)
    asap7sc7p5t_OA_LVT_TT_nldm_211120 (File: /home/v72180/mydata/asic_labs/Lab3/Lab3_modified/lib/db/asap7sc7p5t_OA_LVT_TT_nldm_211120.db)
    asap7sc7p5t_SEQ_SLVT_TT_nldm_220123 (File: /home/v72180/mydata/asic_labs/Lab3/Lab3_modified/lib/db/
asap7sc7p5t_SEQ_SLVT_TT_nldm_220123.db)
Loading file "/home/v72180/mydata/asic_labs/Lab3/Lab3_modified/reports/MIPS.power"
Plain Text  Tab Width: 8  Ln 1, Col 1  INS

```

Figure 6 – Mips.power contents

Checkpoint 4:

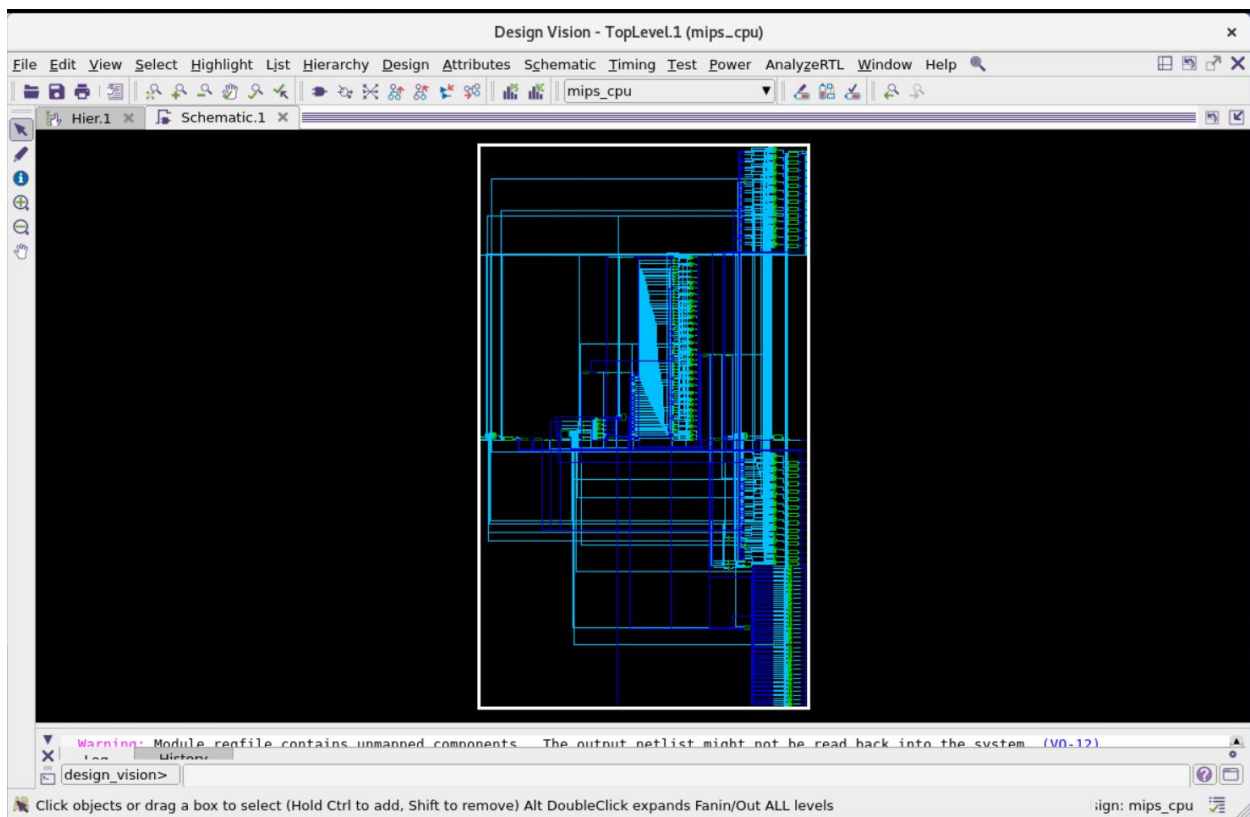


Figure 6 – MIPS CPU Schematic

MIPS timing			
u_sign_extend/out[14] (sign_extend)	0.00	37.89	r
U204/Y (INVx6 ASAP7_75t SL)	2.27	40.16	f
U202/Y (OAI22x1 ASAP7_75t SL)	13.51	53.67	r
u_alu/b[14] (alu)	0.00	53.67	r
u_alu/sub_19/B[14] (alu DW01 sub 2)	0.00	53.67	r
u_alu/sub_19/U13/Y (NAND2xp5 ASAP7_75t SL)	6.24	59.92	f
u_alu/sub_19/U7/Y (AND3x1 ASAP7_75t SL)	14.53	74.45	f
u_alu/sub_19/U5/Y (NAND2xp33 ASAP7_75t SL)	10.63	85.08	r
u_alu/sub_19/U8/Y (OR2x2 ASAP7_75t SL)	14.78	99.86	r
u_alu/sub_19/U31/Y (AND3x1 ASAP7_75t SL)	18.08	117.94	r
u_alu/sub_19/U17/Y (NAND4xp75 ASAP7_75t SL)	14.00	131.94	f
u_alu/sub_19/U05/Y (AOI21x2 ASAP7_75t SL)	20.16	152.11	f
u_alu/sub_19/U63/Y (AOI21xp5 ASAP7_75t SL)	11.91	164.01	r
u_alu/sub_19/U61/Y (XOR2xp5 ASAP7_75t SL)	15.69	179.70	r
u_alu/sub_19/DIFF[28] (alu DW01 sub 2)	0.00	179.70	r
u_alu/U18/Y (NAND2xp5 ASAP7_75t SL)	10.41	190.11	f
u_alu/U6/Y (NAND3x1 ASAP7_75t SL)	17.43	207.54	r
u_alu/U3/Y (NOR4xp75 ASAP7_75t SL)	7.73	215.27	f
u_alu/U8/Y (NAND5xp2 ASAP7_75t SL)	13.19	228.45	r
u_alu/U21/Y (NOR3x1 ASAP7_75t SL)	18.47	246.93	f
u_alu/zero (alu)	0.00	246.93	f
U210/Y (BUFx6f ASAP7_75t SL)	22.10	269.03	f
U331/Y (AOI332xp33 ASAP7_75t SL)	24.32	293.35	r
pc_q_reg[2]/D (DFFASRHONx1 ASAP7_75t SL)	0.00	293.35	r
data arrival time		293.35	
clock clk (rise edge)	200.00	200.00	
clock network delay (ideal)	0.00	200.00	
pc_q_reg[2]/CLK (DFFASRHONx1 ASAP7_75t SL)	0.00	200.00	r
library setup time	-17.19	182.81	
data required time		182.81	
data required time		182.81	
data arrival time		293.35	
slack (VIOLATED)		-110.54	

Figure 7 – Timing report showing slack violation. Lab3_mips_top.v hasn't been modified and Mohammad Mansour checked my scripts and couldn't find a problem